CLINICAL · APPROACH TO

Approach to a runny nose

A runny nose is usually rhinitis, and rhinitis is usually treatable. The rhinologist's task is to make the first and most important distinction: **bilateral** rhinorrhoea is almost always mucosal disease, allergic, non-allergic or infective, while **unilateral** clear rhinorrhoea is a cerebrospinal fluid leak until proven otherwise, and unilateral discoloured or bloody discharge is a foreign body or a tumour until the nose has been examined with an endoscope and, if needed, scanned.

AT A GLANCE

- Clear and watery from one side: test for CSF.
- Unilateral purulent in a child: foreign body.
- Gustatory rhinorrhoea is cholinergic: ipratropium.
- Rhinitis medicamentosa: stop the decongestant.

1 CHAPTER ONE Which side, and what colour?

The first two questions.

1.1 Sorting the discharge

RHINORRHOEA BY SIDE AND CHARACTER

PATTERN	THINK OF	KEY FEATURES
Bilateral, watery, with sneezing and itch	Allergic rhinitis	Seasonal or perennial triggers, itchy eyes, atopy
Bilateral, watery, no itch	Non-allergic rhinitis	Triggered by cold air, smells, food, alcohol; older patients
Bilateral, thick, discoloured	Acute or chronic rhinosinusitis	Blockage, pain, smell loss, pus in the middle meatus
Unilateral, clear, watery, salty, positional	CSF leak	After trauma or surgery, or spontaneous; worse bending forward

PATTERN	THINK OF	KEY FEATURES
Unilateral, foul, purulent	Foreign body (child), dental sinusitis, fungal ball	Smell noticed by others; dental history
Unilateral, blood-stained	Tumour	Obstruction, pain, numbness, proptosis

DO NOT MISS

- **Clear, watery, unilateral discharge**, especially after head injury or sinus surgery, worse on bending forward or straining, or with a salty taste: collect it for beta-2 transferrin or beta-trace protein.
- **Any history of meningitis** in a patient with a runny nose: a CSF leak until proven otherwise.
- **Blood-stained unilateral discharge in an adult:** endoscopy and imaging for a tumour.
- **Unilateral foul discharge in a child:** foreign body.

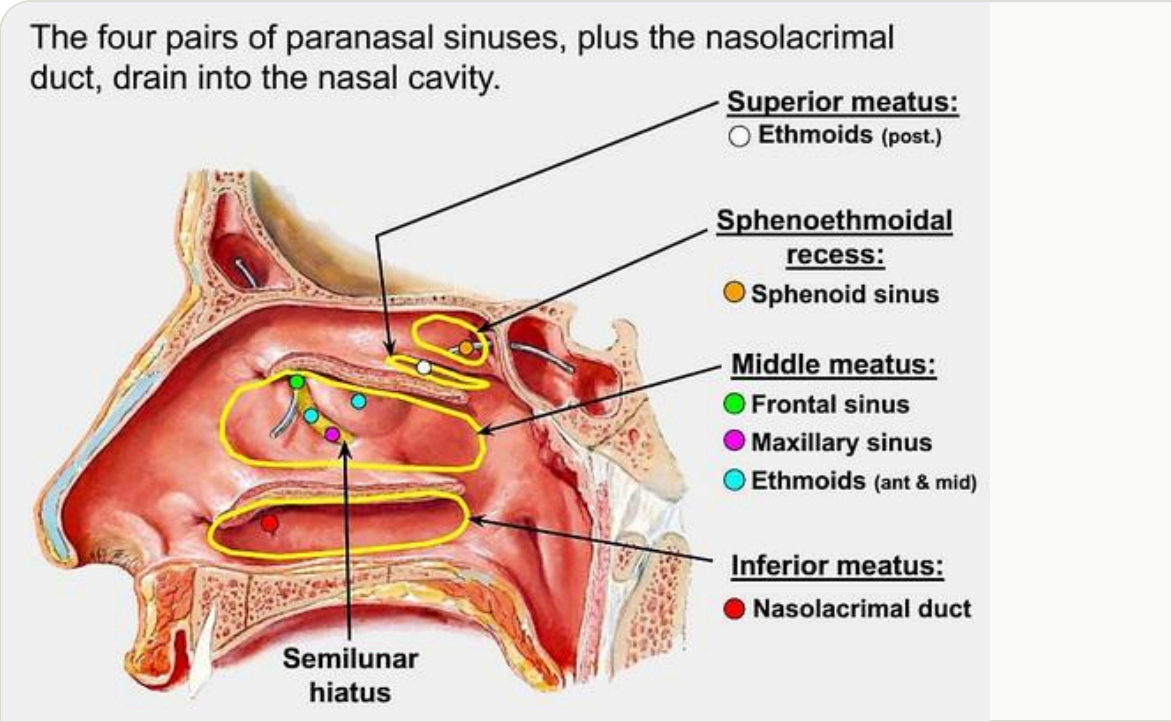


FIGURE 1 Where each sinus and the nasolacrimal duct drains into the meatuses: the site of discharge on endoscopy points to its source.

Anatomy atlas · ENT Steps p. 101

2 CHAPTER TWO Bilateral rhinorrhoea

Mostly mucosal.

2.1 Allergic rhinitis

Allergic rhinitis is an IgE-mediated inflammation producing sneezing, watery discharge, itch of the nose and eyes, and blockage. The **ARIA** classification divides it by duration into *intermittent* (fewer than four days a week or for less than four consecutive weeks) and *persistent*, and by severity into *mild* and *moderate to severe*, depending on effects on sleep, daily activities, work and troublesome symptoms. It is diagnosed from the history and confirmed with skin-prick tests or specific IgE. A subgroup has *local allergic rhinitis*, with typical symptoms and negative skin tests but IgE in the nasal mucosa.

Treatment follows the ARIA steps: allergen avoidance and saline; a non-sedating oral or topical antihistamine for mild intermittent disease; an intranasal steroid, or combined steroid and antihistamine spray, for persistent or moderate to severe disease; and **allergen immunotherapy**, sublingual or subcutaneous, for selected patients whose allergy is confirmed and who do not respond. Leukotriene antagonists help some patients with asthma. Depot and intranasal steroid injections are not recommended.

2.2 Non-allergic rhinitis

Non-allergic rhinitis produces similar symptoms without atopy. It includes **vasomotor** (idiopathic) rhinitis, triggered by temperature change, strong smells and alcohol; **gustatory** rhinitis with eating, especially hot or spicy food; **rhinitis of the elderly**, a profuse clear drip; **hormonal** rhinitis in pregnancy and hypothyroidism; **drug-induced** rhinitis from decongestant overuse, antihypertensives, the oral contraceptive pill, aspirin and NSAIDs; **occupational** rhinitis; and **NARES** (non-allergic rhinitis with eosinophilia syndrome), in which nasal eosinophilia without atopy may precede polyps, asthma and aspirin sensitivity.

When watery discharge dominates, **ipratropium bromide** spray, an anticholinergic, used before meals or triggers, is the drug of choice. Intranasal steroids help NARES and congestion. For intractable cases, the parasympathetic supply can be interrupted surgically: **vidian neurectomy** is effective but risks a dry eye, so the more selective **posterior nasal nerve** procedures (neurectomy, cryoablation or radiofrequency) are now preferred.

2.3 Infective rhinitis and rhinosinusitis

The common cold is the commonest cause of all and settles in about ten days. Discoloured discharge with blockage, facial pain or smell loss for more than twelve weeks is chronic rhinosinusitis and is managed on its own pathway.

3 CHAPTER THREE Unilateral rhinorrhoea

Leak or lesion?

3.1 CSF rhinorrhoea

CSF rhinorrhoea is a communication between the subarachnoid space and the nose. About 80 to 90% are **traumatic**, from head injury or surgery; the rest are **non-traumatic**, and many of these are *spontaneous* leaks in women with obesity and raised intracranial pressure (idiopathic intracranial hypertension), often through the cribriform plate or a lateral sphenoid recess. Spontaneous leaks have the highest recurrence rates.

The discharge is clear, watery and unilateral, worse on bending forward (the *teapot sign*) or straining, and may run backwards when lying down. It does not cause sneezing or itchy eyes. Diagnosis is confirmed by **beta-2 transferrin** or **beta-trace protein** in the fluid; the "halo" and glucose tests are unreliable. **High-resolution CT** shows the bony defect, MRI shows any encephalocele or herniated meninges, and CT cisternography or intrathecal fluorescein at surgery helps find small or intermittent leaks.

Traumatic leaks are first treated conservatively, with bed rest, head elevation, stool softeners and avoidance of nose blowing, and a lumbar drain if needed; many close within a week. Persistent and spontaneous leaks are repaired endoscopically with a graft or a nasoseptal flap, and spontaneous leaks need their raised pressure treated to prevent recurrence. Pneumococcal vaccination is recommended.

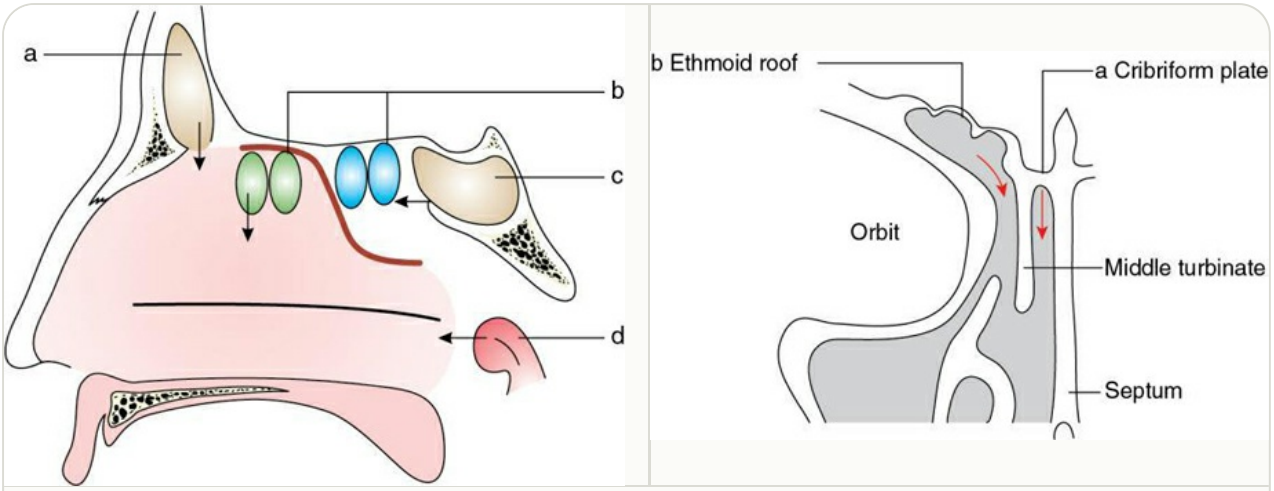


FIGURE 2 Sites of CSF leakage into the nose and ear.
Dhingra, figs 29.4 and 29.5

3.2 Other unilateral causes

A **foreign body** in a child, or after dental or sinus surgery in an adult, gives a foul unilateral discharge. **Dental (odontogenic) sinusitis** and a **fungal ball** affect one maxillary sinus, often with a foul smell. **Choanal atresia** occasionally presents with unilateral discharge. **Tumours**, inverted papilloma and sinonasal malignancy, produce discharge with blockage, bleeding and pain. Unilateral watery discharge with tearing during attacks of severe orbital pain is **cluster headache**.

4 CHAPTER FOUR Pathway

Discharge to diagnosis.

THE RUNNY NOSE

1 Which side?

→ **bilateral**
step 2

→ **unilateral**
step 3

2 Bilateral — itch, sneezing, triggers?

→ **atopic, itchy**
allergic rhinitis — avoidance, antihistamine, steroid spray, immunotherapy

→ **watery, triggered by cold, food or age**
non-allergic — ipratropium

→ **discoloured with smell loss over 12 weeks**
chronic rhinosinusitis

→ **decongestant overuse**
stop it

3 Unilateral — clear and watery?

→ **yes**
collect for beta-2 transferrin, CT for defect, MRI

→ **foul**
endoscopy, remove foreign body, dental review, CT

→ **bloody**
endoscopy and imaging, biopsy

4 CSF leak confirmed

→ **traumatic**
conservative first, repair if persisting

→ **spontaneous**
endoscopic repair and treat raised pressure

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Unilateral clear rhinorrhoea is CSF until proven otherwise;** beta-2 transferrin.
- **The ARIA classification** and stepwise treatment of allergic rhinitis.
- **Non-allergic rhinitis subtypes,** NARES and ipratropium.
- **Vidian and posterior nasal nerve surgery:** what they do and the dry eye risk.
- **Spontaneous CSF leak** and idiopathic intracranial hypertension.

5.2 Common pitfalls

COMMON PITFALLS

- Treating a unilateral watery discharge as rhinitis for months.
- Relying on glucose testing to diagnose CSF.
- Giving steroid sprays for rhinorrhoea in the elderly, where ipratropium works better.
- Repairing a spontaneous leak without addressing raised intracranial pressure.

Written for this app from Oxford Textbook of Otolaryngology ch. 29, 77, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 28. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

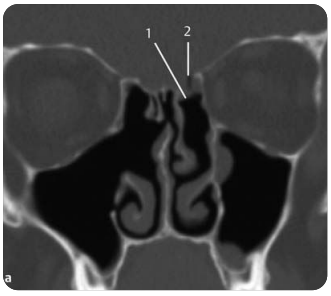
On imaging

The imaging of this topic: key findings, differentials and representative scans.

PARANASAL SINUSES · AFTER SURGERY

Dural Defect after Sinonasal Surgery

- The commonest site of iatrogenic skull base injury is the lateral lamella of the cribriform plate (thinnest bone, where the anterior ethmoidal artery enters), then the posterior ethmoid roof.
- Locate the defect with thin-section coronal CT; MR cisternography (heavily T2-weighted) shows CSF tracking through it; beta-2 transferrin confirms leaked fluid.
- Small defects are repaired endoscopically with grafts; larger ones with a vascularised (nasoseptal) flap.



CT, coronal. This slice at the level of the olfactory fossa shows a bone defect involving the left cribriform plate (1), indicating the location of th…

NOSE & NASAL CAVITY · CONGENITAL

Meningoencephalocele

IMAGING FINDINGS

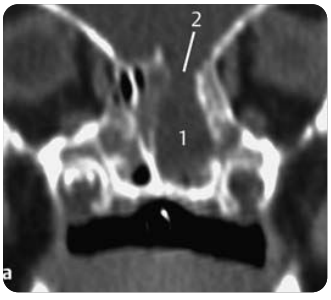
- Bony defect in the cribriform plate, lateral lamella, ethmoid roof or sphenoid
- Soft tissue or fluid through the defect
- MRI: CSF, meninges or brain in continuity with the intracranial contents

TOP DIFFERENTIALS

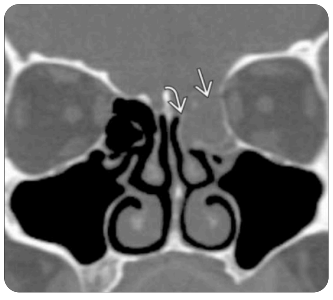
- Skull base defect without leak
- Sinus opacification from mucus

CHECKLIST

- Lateral sphenoid recess and multiple defects
- Signs of raised intracranial pressure: empty sella



CT, coronal. The left nasal cavity is filled with a soft-tissue mass (1) that seems to be connected to brain through a defect in the roof of the…



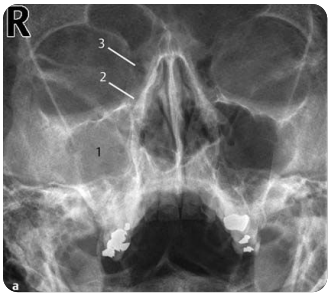
Coronal bone CT shows large bony defect in left ethmoid roof, lateral to insertion of middle turbinate. Because there is complete opacification of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Odontogenic Sinusitis

- Destruction of bony outlines with infiltrative characteristics is suggestive of inverted papilloma, malignancies, or fungal infections.
- Calcifications in the opacified area are suggestive of fungal infections.
- History of frequent dental consultations and resorption of bone around tooth roots may indicate odontogenic infections.
- Long-term antibiotics are an essential part of postoperative treatment.



Plain radiograph, Waters view. There is opacification of the right maxillary sinus (1) as well as in the region of the right ethmoid (2). The right frontal…